

Searching From Inside Files

Sometimes you have a LOT of information *in text files*; so much that you can NOT read through the entire thing. OR sometimes you only want certain information like COUNT or line numbers.

If you did the LAST PDF *you already have the dataset*, if not:
<https://pastebin.com/t7EpMV5J>

Change directories into this Directory `cd search_practice`

Let's say we have a file, and *we want to find* occurrences of "coffee"

> `grep "coffee" coffee_notes.txt` ; `grep` is a tool kind of like CTRL+F, but *outside the file*. So this finds any instance of `coffee` in `coffee_notes.txt`

```
vboxuser@ExampleName:~/search_practice$ grep "coffee" coffee_notes.txt
Espresso: Fine ground coffee
```

Yet, if we LOOK inside the file `nano coffee_notes.txt` there is MORE than one occurrence of "coffee". This is because of **case sensitivity**.

```
Coffee Recipes

Espresso: Fine ground coffee
Latte: Espresso + milk
Americano: Espresso + water
Iced Coffee: Coffee + ice

Coffee Tips:
- Use filtered water
- Grind fresh beans
- Proper temperature
```

We can TURN off *case sensitivity* with `-i`

> `grep -i "coffee" coffee_notes.txt` Filters with `-i` for ANY occurrence of coffee *no matter uppercase or lowercase*, and **spits it out**.

```
vboxuser@ExampleName:~/search_practice$ grep -i "coffee" coffee_notes.txt
Coffee Recipes
Espresso: Fine ground coffee
Iced Coffee: Coffee + ice
Coffee Tips:
```

What if we want to know WHERE in the file it is?

> `grep -n -i "coffee" coffee_notes.txt` This filters for any occurrence of coffee, the option for disabling CASE SENSITIVITY from before is still there, but NOW we have `-n` which also gives the LINE numbers.

```
vboxuser@ExampleName:~/search_practice$ grep -n -i "coffee" coffee_notes.txt
1:Coffee Recipes
3:Espresso: Fine ground coffee
6:Iced Coffee: Coffee + ice
8:Coffee Tips:
```

You can actually COMBINE options AKA flags, `-n -i` can be `-ni`

Let's say we want to search through whole FOLDER, for ANY occurrence of a phrase because we DON'T even know the file? Or its in MORE than one file?

> `grep -n -i -r "coffee" ./` This combines the previous options, with the new `-r` MEANING, search recursively; as in EVERYTHING from `./` (current directory) and below.

```
vboxuser@ExampleName:~/search_practice$ grep -n -i -r "coffee" ./
./shopping.txt:3:COFFEE beans
./shopping.txt:6:Coffee filters
./shopping.txt:7:Coffee mug
./budget.txt:4:Coffee: $45
./budget.txt:7:Coffee: $38
./budget.txt:11:Coffee: $42
./budget.txt:13:Coffee: $35
./budget.txt:15:Coffee: $40
./budget.txt:20:Coffee: $44
./budget.txt:21:Coffee shop: $28
./budget.txt:26:Coffee: $24
./budget.txt:28:Total coffee spending: $296
./passwords.txt:3:Email: CoffeeLover123
./system.log:4:ERROR: Coffee machine offline
./system.log:5>User downloaded coffee_recipes.pdf
./old_taxes.pdf:2:Coffee business expense: $600
./todo.txt:2:1. Buy coffee beans
./todo.txt:6:5. Check bank for coffee charges
./coffee_notes.txt:1:Coffee Recipes
./coffee_notes.txt:3:Espresso: Fine ground coffee
./coffee_notes.txt:6:Iced Coffee: Coffee + ice
./coffee_notes.txt:8:Coffee Tips:
```

Now what if *ON TOP* of that; we want to COUNT the number of OCCURRENCES?

Taking in our old COMMAND; let's have it COUNT the amount of lines with "coffee"

> `grep -i -r "coffee" ./ | wc -l` This is a normal command from the pages before, but we added a "`|`" which *means to take what's on the left, and do whats on the right to it, in this case do a word count.* `wc -l` means to give the count of lines where the phrase is said.

```
vboxuser@ExampleName:~/search_practice$ grep -i -r "coffee" ./ | wc -l
22
```

But, one of these lines have "coffee" more than once, is that counted? *no.*

> `grep -i -r -o "coffee" ./ | wc -l` This `-o` counts more if a word is in a line more than once.

```
vboxuser@ExampleName:~/search_practice$ grep -i -r -o "coffee" ./ | wc -l
23
```

That is ALL you really need for *searching in & out* of Files.